

FIITJEE

ALL INDIA TEST SERIES

FULL TEST – IX

JEE (Main)-2022

TEST DATE: 03-04-2022

ANSWERS, HINTS & SOLUTIONS

Physics

PART – A

SECTION – A

1. C
Sol. Value of masses after explosion

$$\frac{2m}{5}, \frac{2m}{5}, \frac{m}{5}$$

$$P_i = P_f$$

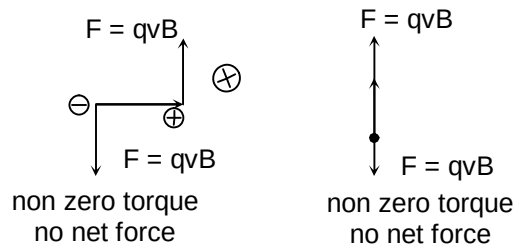
$$0 = \frac{2m}{5}v\hat{i} + \frac{2m}{5}v\hat{j} + \frac{m}{5}\vec{v}'$$

$$|\vec{v}'| = v\sqrt{5}$$

2. C
Sol. Due to formation of depletion layer electric field is setup from N side to P side.

3. C
Sol. Intensity in electromagnetic wave is given by $\frac{1}{2}\epsilon_0 E_0^2 c$ where E_0 is maximum electric field.

4. C
Sol. Since q_{net} on dipole = 0 $\Rightarrow$ no net force



5. C

Sol. As shown, we conclude that A and C are continuous function while B is discrete function.

6. A

 Sol. From A : $\Delta Q > 0$, $\Delta U = 0$, $\Delta W > 0$

 From B : $\Delta Q < 0$, $\Delta U = 0$, $\Delta W < 0$

7. A

 Sol. The number for yellow colour is 4. The number for violet colour is 7. Brown colour gives a multiplier of 10^1 . Gold indicates a tolerance of 5%. So the resistance of the given resistor is $47 \times 10\Omega \pm 5\%$

8. C

 Sol. $f = R/2$

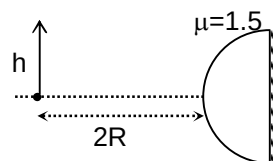
9. D

Sol. Here, three optical phenomena take place first refraction, then reflection, and finally refraction.

For refraction at first

$$\frac{1.5}{v} - \frac{1}{-2R} = \frac{1.5-1}{R} \Rightarrow \frac{1.5}{v} = 0 \Rightarrow v = \infty$$

i.e., rays after refraction get parallel to axis and strike the mirror normally, get retraced back and the final image is formed at the same place where the object is and of the same size. Image would be real.



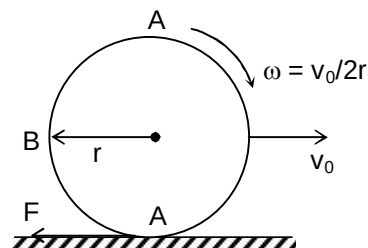
10. A

Sol. Angular momentum can be conserved about lowest point

$$mv_0r + \frac{2}{5}mr^2 \frac{v_0}{2r} = mv'r + \frac{2}{5}mr^2 \frac{v'}{r}$$

$$\frac{10mv_0r + 2Mv_0r}{10} = \frac{7mv'r}{5}$$

$$v' = \frac{6v_0}{7}$$



11. B

Sol. Centre of mass ring must be vertically below point A.

12. C

Sol. Let initially they have n atoms

A → the half time two hours

B → the half time four hours

 After two hours A possesses 1 half life and B possesses $\frac{1}{2}$ half life

$$\text{Remaining atoms of A} = \frac{n}{2^1}$$

$$\text{Remaining atoms of B} = \frac{n}{(2)^{1/2}}$$

13.

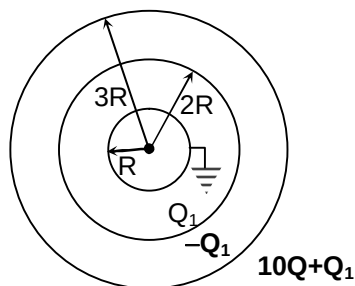
C

Sol. $\frac{Ak_1(T_1 - T)}{\ell_1} = \frac{k_2(T - T_2)A}{\ell_2}$

14.

A

Sol. $\frac{KQ}{R} - \frac{KQ_1}{2R} + \frac{K(10Q + Q_1)}{3R} = 0$
 $3Q_1 + 20Q + 2Q_1 = 0 \Rightarrow Q_1 = -4Q$



15.

B

Sol. $W_A > W_B$ as mass of water in A is more than in B

$$P_A = P_B$$

Area of A = Area of B

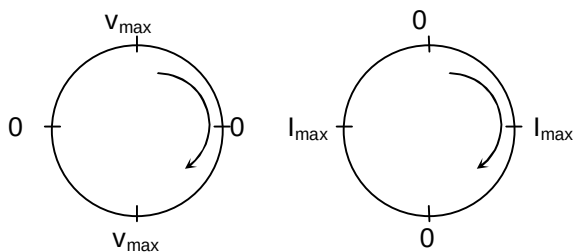
$$\text{or } P_A \text{ Area}_A = P_B \text{ Area}_B$$

$$\text{or } F_A = F_B$$

16.

A

Sol. $\Rightarrow$ Current lags voltage by 90°



17.

A

Sol. Path difference $\Delta x = 2nt - \frac{\lambda}{2}$

For maximum $\frac{8}{3}t - \frac{\lambda}{2} = n\lambda$

$$\frac{8}{3}t = \left(n + \frac{1}{2}\right)630 \quad \dots(i)$$

For minimum $\frac{8}{3}t - \frac{\lambda}{2} = \left(m - \frac{1}{2}\right)\lambda$

$$\frac{8}{3}t = m(525) \quad \dots(ii)$$

Equation (i) divided by equation (ii)

$$\frac{n + \frac{1}{2}}{m} = \frac{2.5}{3}$$

i.e. $n + \frac{1}{2} = 2.5$

So from (i) $t = 590.625$

18. A

 Sol. At the open end pressure node is present so that at $x = 0$ p should be zero and $v = f\lambda$,

 For first overtone $\frac{3\lambda}{4} = 0.80$

$$\Rightarrow f = 300$$

$$\Rightarrow \omega = 2\pi f = 600\pi$$

$$\text{Hence } P = P_0 \sin\left(\frac{15\pi x}{8}\right) \cos(600\pi t)$$

19. A

 Sol. Ist Case:

$$i = \frac{20}{R+2}$$

Reading of voltmeter

$$V = iR = 10 \text{ Volt}$$

$$\Rightarrow \frac{20R}{R+2} = 10 \Rightarrow 2R = R+2$$

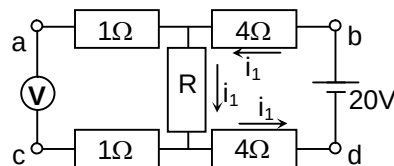
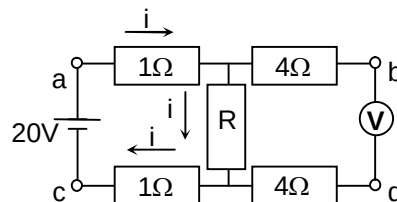
$$R = 2\Omega$$

 IInd Case:

$$i_1 = \frac{20}{4+4+2} = 2 \text{ amp}$$

Reading of voltmeter

$$V = i_1 R = 4 \text{ Volt}$$



20. D

 Sol. Current through battery is $\frac{3}{2} = 1.5$ ampere.

SECTION – B

21. 00001.10

 Sol. Simple pendulum $T = 2\pi\sqrt{\frac{L}{g}} \Rightarrow g = \frac{4\pi^2 L}{T^2}$

$$L = 100 \text{ cm} = 1 \text{ m}, T = 2$$

... (i)

Taking log both side in equation (i)

$$\ln g = \ln 4\pi^2 + \ln L - \ln T$$

$$\frac{dg}{g} = 0 + \frac{dL}{L} - 2\frac{dT}{T}$$

% error

$$\frac{dg}{g} \times 100 = \frac{dL}{L} \times 100 + 2\frac{dT}{T} \times 100$$

$$= \frac{0.1}{100} \times 100 + 2 \times \frac{0.01}{2} \times 100 = 0.1 + 1 = 1.10$$

22. 00002.00

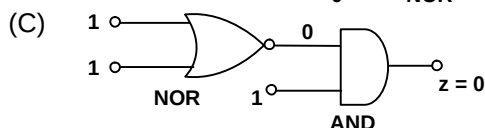
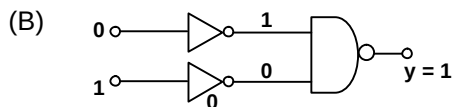
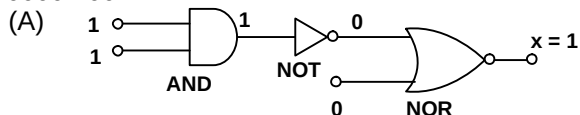
 Sol. $\frac{4}{3}\pi R^3 - 1000 \times \frac{4}{3}\pi r^2$

$$R = 10r$$

$$V = \frac{k(1000q)}{10r}$$

23. 00002.00

Sol.



$$\begin{aligned} x &= 1 \\ y &= 1 \\ z &= 0 \\ \Rightarrow x + y + z &= 2 \end{aligned}$$

24. 00001.50

Sol.

An initial mechanical energy of the block is mgh and final is zero, so loss in mechanical energy = mgh . This mechanical energy is lost in doing work against friction in the flat part, So, loss in mechanical energy = work done against friction

$$\text{Or } mgh = \mu mgd \text{ i.e., } d = \frac{h}{\mu} = \frac{1.5}{0.2} = 7.5 \text{ m}$$

After starting from B, the block will reach C and then will rise up till the remaining KE at C is converted into potential energy. It will then again descend and at C will have the same value as it had when ascending, but now it will move from C to B. The same will be repeated and finally the block will come to rest at E such that

$$BC + CB + BE = 7.5$$

$$\text{or } 3 + 3 + BE = 7.5$$

$$\text{ie. } BE = 1.5$$

So, the block comes to rest at the centre of the flat part.

25. 00000.60

Sol.

$$\frac{n_1(4)}{n_2(32)} = \frac{1}{4} \Rightarrow n_1 = 2n_2$$

$$\text{Now, } C_v = \frac{n_1\left(\frac{3}{2}R\right) + n_2\left(\frac{5}{2}R\right)}{n_1 + n_2} = \frac{11}{6}R$$

$$C_p = C_v + R = \frac{17}{6}R$$

$$\therefore \gamma = \frac{C_p}{C_v} = \frac{17}{11} = 1 + \frac{6}{11}$$

26. 00001.41

Sol.

$$\gamma = 3/2$$

$$V_{\text{rms}} = \sqrt{\frac{3RT}{M}}, \quad V_{\text{sound}} = \sqrt{\frac{\gamma RT}{M}} \Rightarrow \frac{V_{\text{rms}}}{V_{\text{sound}}} = \sqrt{\frac{3}{\gamma}} = \sqrt{2}$$

27. 00000.25

Sol.

In time t heat produced by a DC current I_1 is given by

$$H_1 = I_1^2 R t$$

$$\Rightarrow H_1 = (2)^2 (2)t = 8t$$

RMS value of AC is given as

$$i_{\text{rms}} = \frac{i_0}{\sqrt{2}} = \frac{2}{\sqrt{2}} = \sqrt{2} \text{ A}$$

Heat produced by the rms current in time t is given as

$$H_2 = i_{\text{rms}}^2 R t$$

$$\Rightarrow H_2 = (\sqrt{2})^2 (1)t = 2t$$

$$\Rightarrow \frac{H_1}{H_2} = \frac{8t}{2t} = 4$$

28. 00004.00

Sol. The energy stored in a coil of self inductance L when the current changes by $2A$ is given as

$$E = \frac{1}{2} L I^2 = \frac{1}{2} \times (2) \times (2)^2 = 4J$$

This is equal to the amount of energy spent by the source which is supplying the current.

29. 00000.50

Sol. $L = \frac{nh}{2\pi}$

$$L_{(n+1)} - L_n = (n+1) \frac{h}{2\pi} - \frac{nh}{2\pi} = \frac{h}{2\pi} = (0.5) \frac{h}{\pi}$$

30. 00000.60

Sol. The apparent position of the object O from the surface of water is

$$h' = \frac{h}{\mu'} = \frac{0.4}{4/3} = 0.3$$

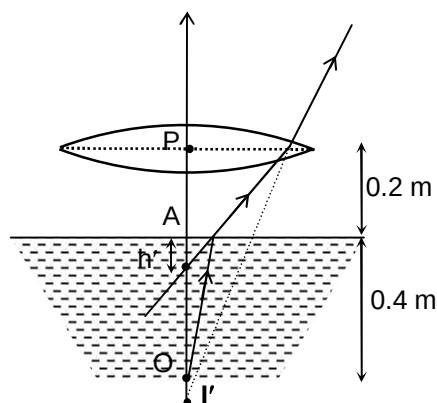
The distance $PI = 0.3 + 0.2 = 0.5 \text{ m}$

For convex lens, we use

$$u = -0.5\text{m}, f = +3\text{m}$$

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}, \text{ we have}$$

$$\Rightarrow \frac{1}{v} - \frac{1}{-0.5} = \frac{1}{3} \Rightarrow v = -0.6 \text{ m}$$

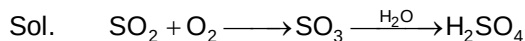


Chemistry

PART – B

SECTION – A

31. D



Composition of London smog is fog of H_2SO_4 droplets deposited on carbon particles. So, it is sulphurous smog.

32. C

Sol. Elastomers – Polymer chains are held together by weak van der Waal's interaction.

Fibres – These have the strongest force of attraction due to formation of H-bonding or dipole-dipole interaction.

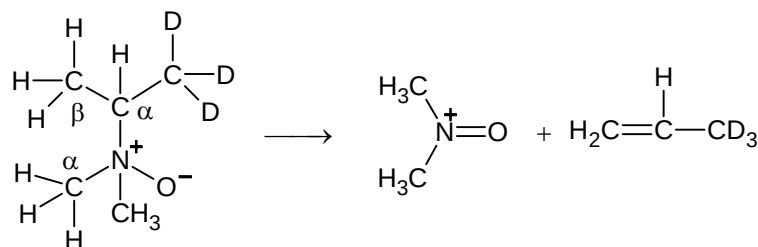
Force of attraction : Fibre > Plastic > Elastomers.

33. B

Sol. Due to the absence of a free hemiacetal or hemiketal group, sucrose does not exhibit mutarotation.

34. D

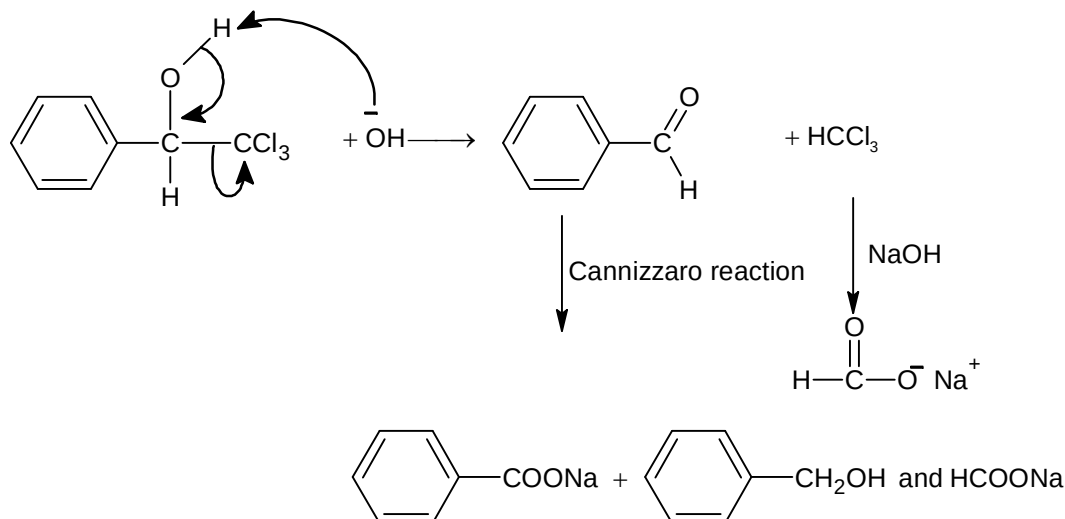
Sol.



C – H bond is weaker than C – D.

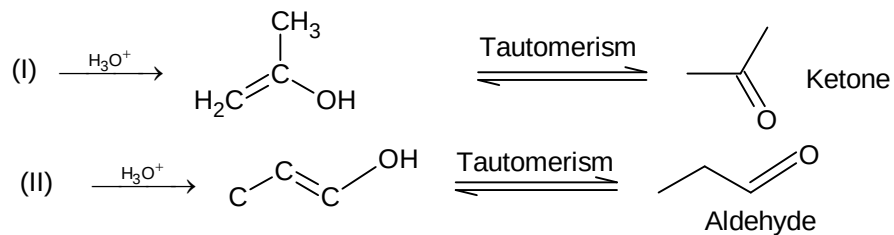
35. D

Sol.



36. A

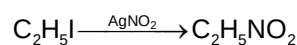
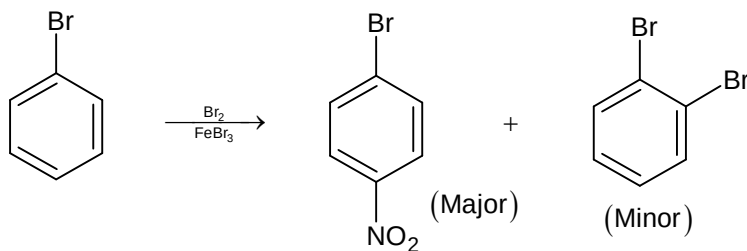
Sol.



Aldehyde and Ketone can be distinguished by Fehling's solution.

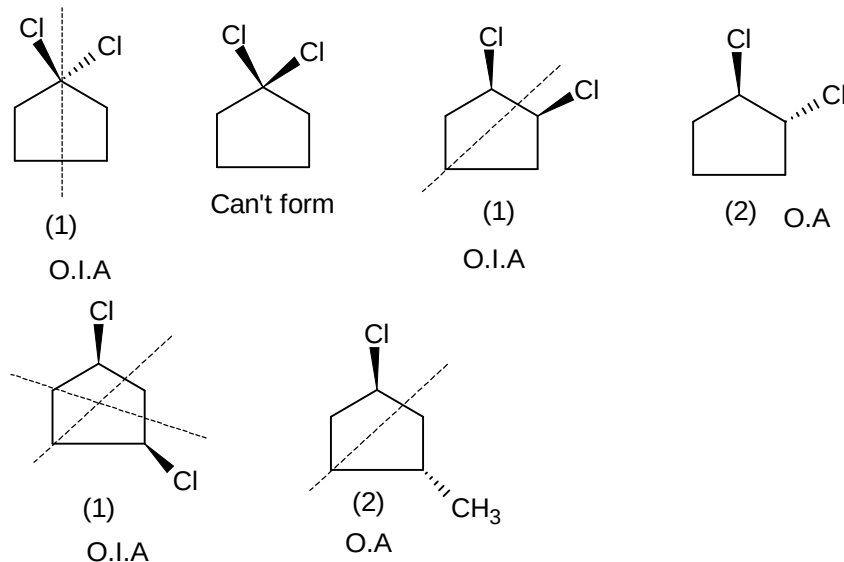
37. A

Sol.


 AgNO_2 is predominantly covalent in nature.


38. D

Sol.



$$T_n = 7$$

39. A

Sol.

 O_2^- (Superoxide) – 1 unpaired electron

 O_2^{2-} (Peroxide) – No unpaired electron

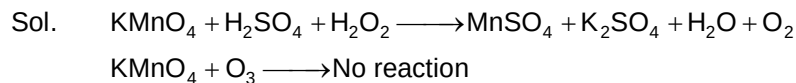
 O^{2-} – No unpaired electron

 NO_2^+ – No unpaired electron

40. C

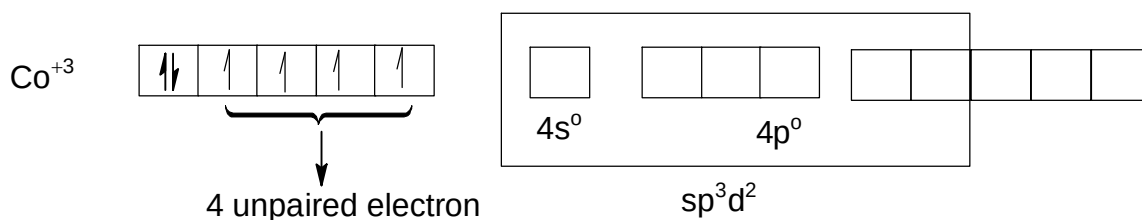
- Sol. 1. $\text{Li} < \text{Na} > \text{K} < \text{Rb} < \text{Cs}$ (Due to abnormal increase in size of K).
 2. $\text{B} < \text{Ga} < \text{Al} < \text{In} < \text{Tl}$ (Ga comes after 3d series).
 3. Due to back bonding.
 4. $\text{LiF} < \text{NaF} < \text{KF} < \text{RbF} < \text{CsF}$ (Due to lattice energy).

41. C

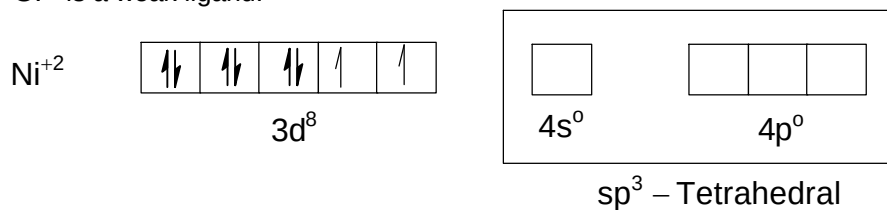


42. A

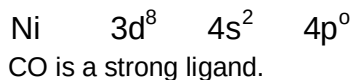
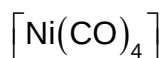
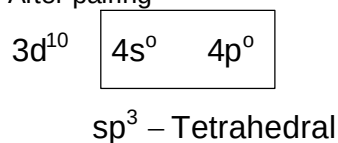
- Sol. (I) $[\text{CoF}_6]^{-3}$
 $\text{Co} \quad 3d^6 \quad 4s^0 \quad 4p^0 \quad 4d^0$
 F^- is a weak ligand.



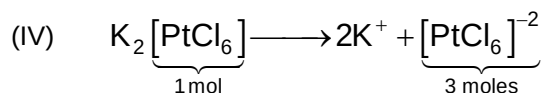
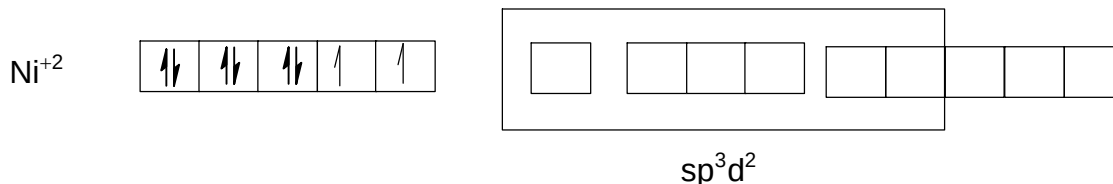
- (II) $[\text{NiCl}_4]^{-2}$
 $\text{Ni}^{+2} \quad 3d^8 \quad 4s^0 \quad 4p^0 \quad 4d^0$
 Cl^- is a weak ligand.



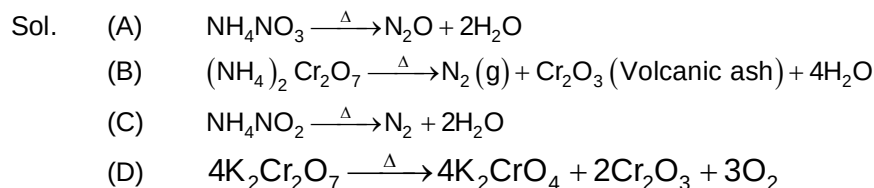
After pairing



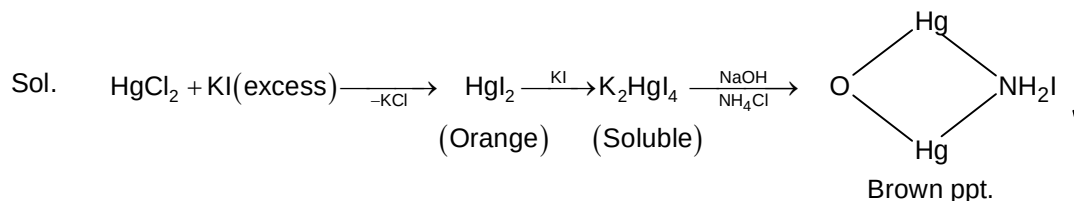
- (III) $[\text{Ni}(\text{NH}_3)_6]^{+2}$
 $\text{Ni}^{+2} \quad 3d^8 \quad 4s^0 \quad 4p^0 \quad 3d^0$



43. B



44. C



45. B

Sol. Final P is same in both which is different from initial $T_1 > T_2$.
Let initial P = P_1

Final $P = P_2$

x is number of moles transfereed from A to B

$$P_2 V = (n - x)RT_1, \quad P_2 V = (n + x)RT_2$$

$$(n-x)RT_1 = (n+x)RT_2$$

$$nT_1 - T_1x = nT_2 + xT_2$$

$$\Rightarrow n(T_1 - T_2) = x(T_2 + T_1)$$

$$X = \frac{n(T_1 - T_2)}{T_1 + T_2}$$

$$V = \frac{nRT_1}{P_1}$$

$$P_2 \times \frac{nRT_1}{P_1} = \left(n - \frac{n(T_1 - T_2)}{T_1 + T_2} \right) RT_1$$

$$P_2 = \left\{ \frac{n(T_1 + T_2) - n(T_1 - T_2)}{T_1 + T_2} \right\} \frac{P_1}{n}$$

$$P_2 = \left(\frac{nT_1 + nT_2 - nT_1 + nT_2}{T_1 + T_2} \right) \frac{P_1}{n}$$

$$P_2 = \frac{2nT_2}{T_1 + T_2} \cdot \frac{P_1}{n}$$

$$P_2 = \frac{2P_1T_2}{T_1 + T_2}$$

46. A

Sol. $\Delta T_f = i \times K_f \times m$ $\Delta T_f \propto \text{Number of particles}$ $\text{NaNO}_3 \longrightarrow 0.10$ $\text{CaCl}_2 \longrightarrow 0.12$ $\text{Glucose} \longrightarrow 0.140$ $\text{ZnSO}_4 \longrightarrow 0.150$ $\Delta T_f \longrightarrow \text{ZnSO}_4 > \text{Glucose} > \text{CaCl}_2 > \text{NaNO}_3$ $T_f \longrightarrow \text{ZnSO}_4 < \text{Glucose} < \text{CaCl}_2 < \text{NaNO}_3$

47. A

Sol. Colour of colloidal solution depends upon the size of the colloidal particles.

48. A

Sol. (A) Packing efficiency in FCC = $\frac{4 \times \frac{4}{3} \pi r^3}{a^3} \times 100 = 74\%$

(B) Electrons are present in anion vacancies.

(C) When Ge is doped with As, then 5th electron of As is free which increases electrical conductivity rapidly.(D) Bragg's equation is $n\lambda = 2d \sin \theta$.

49. D

Sol. $V = \frac{nRT}{P} = \frac{200 \times 0.0821 \times T}{16.42}$ $\log V = \log T$

50. C

Sol. (A) $\text{FeCl}_3 + \text{NaOH} \longrightarrow \text{Fe}(\text{OH})_3 + \text{NaCl}$

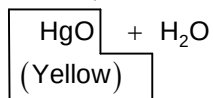
Red ppt.

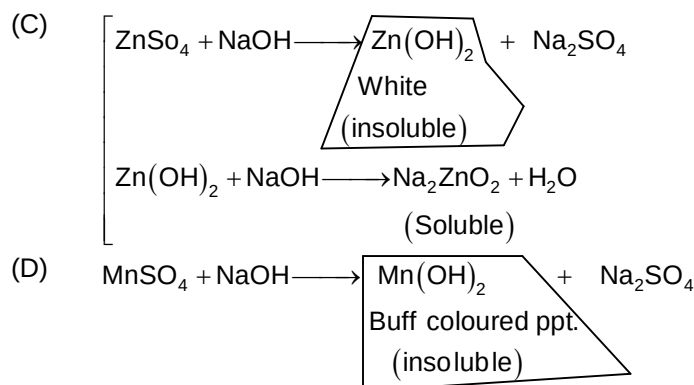
(Insoluble)

(B) $\text{HgCl}_2 + \text{NaOH} \longrightarrow [\text{Hg}(\text{OH})_2] + \text{NaCl}$

unstable

↓





SECTION – B

51. 00103.47

Sol. Saponification number is milligram of KOH or NaOH consumed by 1 g oil.

$$\text{Equivalent of NaOH used by oil} = [25 \times 0.40 - 4 \times 0.28 \times 2] \times 10^{-3}$$

$$= 7.76 \times 10^{-3}$$

$$\text{Moles of NaOH} = 7.76 \times 10^{-3}$$

$$\therefore \text{Mass of NaOH} = 7.76 \times 10^{-3} \times 40 \times 10^3 \text{ mg}$$

$$= 310.4 \text{ mg}$$

$$\therefore \text{Saponification} = \frac{310.4}{3}$$

$$= 103.46666$$

$$= 103.47$$

52. 00019.55

Sol. $E_{\text{photon}} = (13.6) \times 3^2 \left[\frac{1}{2^2} - \frac{1}{4^2} \right]$

$$= 13.6 \times \frac{27}{16}$$

$$\text{Energy required to remove electron from } n = 2 \text{ in H-atom} = 13.6 \times 1^2 \times \left[\frac{1}{4} - \frac{1}{\infty^2} \right]$$

$$= \frac{13.6}{4}$$

$$\therefore \text{Kinetic energy of ejected electron} = 13.6 \times \frac{27}{16} - \frac{13.6}{4}$$

$$= \frac{13.6}{4} \left[\frac{27}{4} - 1 \right] = \frac{13.6}{4} \left(\frac{23}{4} \right)$$

$$= 19.55 \text{ eV}$$

53. 00000.422

Sol. $18\text{g H}_2\text{O} = 6.022 \times 10^{23} \times 2 \text{ atoms of H}$

$$18\text{g H}_2\text{O} \text{ has } {}^3_1\text{H atoms} = 6 \times 10^{-18} \times 6.022 \times 10^{23} \times 2$$

$$\therefore 10\text{g H}_2\text{O} \text{ has } {}^3_1\text{H atoms} = \frac{6 \times 10^{-18} \times 6.022 \times 10^{23} \times 2 \times 10}{18}$$

$$= 2.0073 \times 2 \times 10^6$$

$$\begin{aligned}
 &= 4.0146 \times 10^6 \\
 \frac{0.693}{12.3} &= \frac{2.303}{40} \log \frac{4.015 \times 10^6}{N} \\
 \log \frac{4.015 \times 10^6}{N} &= \frac{0.693 \times 40}{12.3 \times 2.303} \\
 &= 0.978 \\
 \frac{4.015 \times 10^6}{N} &= 9.506 \\
 N &= \frac{4.015 \times 10^6}{9.506} = 0.422 \times 10^6
 \end{aligned}$$

54. 00008.15

Sol. $K_{sp} [\text{Co}(\text{OH})_2] = [\text{Co}^{2+}] [\text{OH}^-]^2 = 2.5 \times 10^{-16}$

$$1.25 \times 10^{-4} \times [\text{OH}^-]^2 = 2.5 \times 10^{-16}$$

$$[\text{OH}^-]^2 = \frac{2.5 \times 10^{-16}}{1.25 \times 10^{-4}} = 2 \times 10^{-12}$$

$$[\text{OH}^-] = 1.414 \times 10^{-6}$$

$$\text{pOH} = -\log 1.414 \times 10^{-6}$$

$$= 6 - 0.1504$$

$$= 5.8496$$

$$\approx 5.85$$

$$\therefore \text{pH} = 14 - 5.85 = 8.15$$

55. 00351.02

Sol. $\text{CaO}(\text{s}) + \text{CO}_2(\text{g}) \longrightarrow \text{CaCO}_3(\text{s})$

$$\Delta H_f^\circ = \Delta H_f^\circ (\text{CaCO}_3, \text{s}) - \Delta H_f^\circ (\text{CaO}, \text{s}) - \Delta H_f^\circ (\text{CO}_2, \text{g})$$

$$= -1207 + 635 + 394$$

$$= -178 \text{ kJ/mol}$$

$$\Delta E = \Delta H - \Delta n_g RT$$

$$= -178 - \frac{(-1) \times 8.3 \times 300}{1000}$$

$$= -178 + 2.49$$

$$= -175.51 \text{ kJ}$$

$$n_{\text{CaO}} = \frac{112}{56} = 2$$

$$q_v = n \Delta E = 2 \times (-175.51)$$

$$= -351.02 \text{ kJ}$$

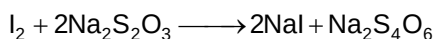
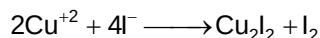
56. 00118.75

Sol. Initial milli moles of $\text{Cu}^{+2} = 50 \times 0.1 = 5$

Meq or m moles of H^+ produced $= 50 \times 10^{-2} = 0.5$

$$\therefore \text{m moles of } \text{Cu}^{+2} \text{ converted into Cu} = \frac{0.5}{2} = 0.25$$

m moles of Cu^{+2} left in solution $= 5 - 0.25 = 4.75$



m moles of Cu^{+2} left = m moles of $\text{Na}_2\text{S}_2\text{O}_3$

$$4.75 = V \times 0.04$$

$$V = \frac{4.75}{0.04} = 118.75 \text{ ml}$$

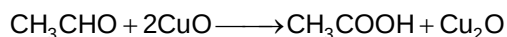
$$= 118.75$$

57. 00031.02

Sol. $W_{\text{C}_2\text{H}_5\text{OH}} + W_{\text{CH}_3\text{CHO}} = 0.535 \text{ g}$

Let x g of $\text{C}_2\text{H}_5\text{OH}$ and y g of CH_3CHO in the mixture

$$x + y = 0.535 \text{ g}$$



$$44 \text{ g} \quad 79.6 \times 2 \quad 63.5 \times 2 + 16$$

Molecular mass of $\text{CuO} = 63.5 + 16 = 79.6$

143 g of Cu_2O is produced by 44 g

$$1.2 \text{ g } \text{Cu}_2\text{O} \text{ is produced by } \frac{44}{143} \times 1.2$$

$$= 0.369$$

$$\therefore W_{\text{C}_2\text{H}_5\text{OH}} = 0.535 - 0.369 = 0.166$$

$$\% \text{ of } \text{C}_2\text{H}_5\text{OH} = \frac{0.166}{0.535} \times 100 = 31.02$$

58. 00001.75

(Range : 1.72 – 1.75)

Sol. $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$

$$5 \quad 0 \quad 0$$

$$5 - x \quad x \quad x$$

$$\text{Total number of moles} = 5 - x + x + x + 4 = 5 + x + 4 = 9 + x$$

$$n_{\text{total}} = \frac{PV}{RT} = \frac{4.678 \times 110}{0.0821 \times 523} = 11.984$$

$$\approx 12$$

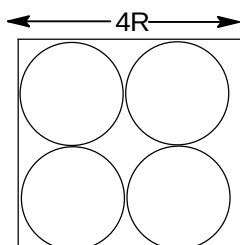
$$x = 11.984 - 9 = 2.984 \approx 3$$

$$K_p = \frac{\left(\frac{3}{12} \times 4.678\right)^2}{\left(\frac{2}{12} \times 4.678\right)} = 1.754$$

59. 00047.60

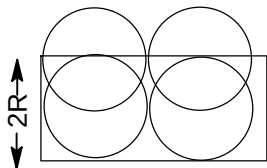
(Range : 47.60 – 47.70)

Sol.



$$\begin{aligned}\text{Volume} &= 4R \times 4R \times 2R \\ &= 32R^3\end{aligned}$$

$$\text{Volume of 4 spheres} = 4 \times \frac{4}{3} \pi R^3$$



$$\% \text{ vacant space} = 100 - 100 \times \frac{16\pi R^3}{3 \times 32R^3}$$

$$\text{With, } \pi = \frac{22}{7}$$

$$= 100 \left(1 - \frac{22}{7 \times 6} \right)$$

$$= 100 \left(1 - \frac{22}{42} \right)$$

$$= 100 \left(1 - \frac{11}{21} \right)$$

$$= 100 \left(\frac{21-11}{21} \right)$$

$$= 100 \times \frac{10}{21}$$

$$= 47.62\%$$

$$\text{With, } \pi = 3.14$$

$$= 100 - 100 \times \frac{\pi}{6}$$

$$= 100 \left(1 - \frac{\pi}{6} \right)$$

$$= 100(1 - 0.523)$$

$$= 47.7\%$$

$$\text{Range : 47.6 to 47.7}$$

60. 00023.39

$$\text{Sol. } 150.7(X) + 52.8(1-X) = +75.7$$

$$\Rightarrow 150.7X + 52.8 - 52.8X = 75.7$$

$$\Rightarrow 97.9X = 22.9$$

$$\Rightarrow X = 22.9 \div 97.9 = 0.23391$$

$$= 0.23391$$

$$\% \text{ of } X = 23.39\%$$

Mathematics

PART – C

SECTION – A

61. B

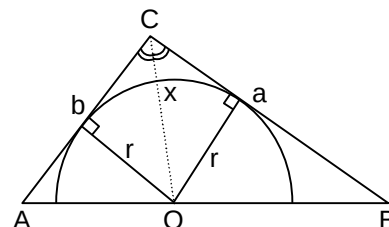
 Sol. Put $1 + xe^{-\sin x} = t$

62. C

 Sol. $\frac{1}{2}r(a+b) = \frac{1}{2}ab \sin C$

$$r = \frac{ab}{a+b} \sin C ; x = \frac{2ab}{a+b} \cos \frac{C}{2}$$

$$\frac{r}{x} = \frac{2 \sin \frac{C}{2} \cos \frac{C}{2}}{2 \cos \frac{C}{2}} = 8 \Rightarrow \sin \frac{C}{2} = \frac{1}{2} \Rightarrow C = \frac{\pi}{3}$$



63. B

 Sol. Total number of matrices = ${}^{11}C_3 - 4^6 C_3 = 85$

 Number of skew symmetric matrix ($a_{ii} = 0$), $a_{ij} (i \neq j) = \pm 1, \pm 2, 0$

64. A

Sol. Use graph

65. C

 Sol. $n(E^c) = {}^3C_1 4! 2! - {}^3C_2 3! (2!)^2 + {}^3C_3 (2!)^4 = 88$

$$\therefore P(E) = 1 - P(E^c) = 1 - \frac{88}{120} = \frac{4}{15}$$

66. C

 Sol. $\cos x d(\ln \sin y) + \ln \sin y d(\cos x) + \sin x d(\ln \cos y) + \ln \cos y d(\sin x) = 0$
 $\Rightarrow d(\cos x \ln \sin y) + d(\sin x \ln \cos y) = 0 \Rightarrow \cos x \ln (\sin y) + \sin x \ln (\cos y) = C$

$$C = \ln \frac{1}{\sqrt{2}}. \text{ Also, } \ln(\sin y) = \ln \frac{1}{\sqrt{2}} \Rightarrow y = \frac{\pi}{4}$$

67. D

$$\text{Sol. } \int_{-1}^0 (e^x - \ln(x+2)) dx + \int_0^{\frac{e-2}{2}} (\ln(e-x) - \ln(x+2)) dx = 1 - \frac{1}{e} + \int_{\frac{e+2}{2}}^e \ln t dt + \int_1^{\frac{e+2}{2}} \ln t dt$$

$$1 - \frac{1}{e} + e + 1 - \ln\left(\frac{e+2}{2}\right) = 2 + e - \frac{1}{e} - (e+2) \ln\left(\frac{e+2}{2}\right)$$

68. B

$$\text{Sol. } \frac{r(r+1)}{(r+1)!} - \frac{(r+1-r)}{(r+1)!} + \frac{(r+1-1)}{(r+1)!}$$

$$T_r = \frac{r}{r!} - \frac{(r+1)}{(r+1)!} + 2\left(\frac{r+1}{(r+1)!} - \frac{(r+2)}{(r+2)!}\right)$$

$$\sum_{r=1}^n T_r = 3 - \left(\frac{n+3}{(n+1)!}\right)$$

69. A

Sol. $x - y + \sin \alpha \cdot z + \lambda(x - y - z) = 0$ (1)

$$x + \sin \alpha \cdot y - z = 0$$

$$\Rightarrow \frac{1+\lambda}{1} = -\frac{(\lambda+1)}{\sin \alpha} = \frac{\sin \alpha - \lambda}{-1}$$

$$\Rightarrow \sin \alpha = -1$$

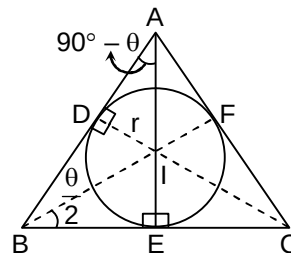
70. C

Sol. $P = r \left(4 \cot \frac{\theta}{2} + 2 \tan \theta \right)$

$$\frac{dp}{d\theta} = 0$$

$$\Rightarrow \tan \frac{\theta}{2} = \frac{1}{\sqrt{3}} ; \theta = \frac{\pi}{3}$$

$$\therefore P_{\min} = 6\sqrt{3}r$$



71. D

Sol. Corrected mean = $\frac{800 - 40 + 60}{20} = 41$

$$\text{Incorrect SD} = \sqrt{\frac{\sum x_i^2}{20} - \left(\frac{\sum x_i}{n} \right)^2}$$

$$\Rightarrow \sum x_i^2 = 20[225 + 1600] = 36500$$

$$\Rightarrow \text{Corrected sum} = 36500 - 40^2 + 60^2 = 38500$$

$$\therefore \text{Variance} = \frac{38500}{20} - (41)^2 = 1925 - 1681 = 244$$

72. C

Sol. Use truth table

73. A

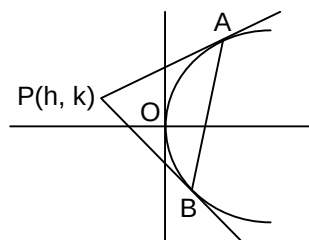
Sol. Equation of AB is $yk = 2(x + h)$
Combined equation of OA and OB is given by

$$y^2 = 4x \left(\frac{yk - 2x}{2h} \right)$$

$$\text{i.e., } 4x^2 + hy^2 - 2kxy = 0 \quad \dots (1)$$

$$x^2 - 7cy^2 - 2cxy = 0 \quad \dots (2)$$

$$\frac{4}{1} = \frac{h}{-7c} = \frac{-2k}{-2c} \therefore \text{locus of P is } y = -\frac{x}{7}$$



74. B

Sol. $1 - \frac{b^2}{a^2} = 1 - \frac{8}{a^2} \Rightarrow b^2 = 8$

$$\text{Area of parallelogram} = |2a \cos \theta - 2\sqrt{2} \sin \theta| \cdot \frac{2\sqrt{2}a}{\sqrt{a^2 + 2}}$$

$$A_{\max} = 4\sqrt{2}a = 16\sqrt{2} \Rightarrow a = 4$$

75. C

Sol. If $(h, 0)$ be the mid-point of chord, then $h^2 - 3\sqrt{2}\sin\theta h + 4\sin^2\theta + \frac{\cos^2\theta}{6} = 0$

For non real root $D < 0 \Rightarrow \tan^2\theta < \frac{1}{3}$

76. D

Sol. $y - 0 = t(x - a)$ (1)

$x = at^2$ (2)

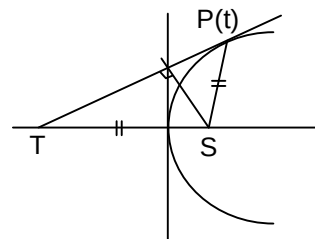
Locus of orthocentre is $\left(\frac{y}{x-1}\right)^2 = \frac{x}{a}$

$\Rightarrow x(x-a)^2 = ay^2$

For intersection $x(x-a)^2 = a \cdot 4ax$

$\Rightarrow x = 0, 3a$

Point of intersection $(3a, \pm 2\sqrt{3}a)$



77. C

Sol. $\hat{i} + 2\hat{j} + \hat{k} = \lambda \left(\frac{\hat{i} + \hat{j}}{\sqrt{2}} + \hat{c} \right) \Rightarrow \hat{c} = \left(\frac{1}{\lambda} - \frac{1}{\sqrt{2}} \right) \hat{i} + \left(\frac{2}{\lambda} - \frac{1}{\sqrt{2}} \right) \hat{j} + \frac{\hat{k}}{\lambda}$

$\Rightarrow \lambda = \sqrt{2} \therefore \hat{c} = \hat{j} + \hat{k}$

78. A

Sol. $\frac{AB}{BC} = 1 + 2\cos 2\theta$

79. D

Sol. $x + x = p \Rightarrow x = \frac{p}{2}$; $x + y = p \Rightarrow y + x = p$; $x + y = p, y + z = p$, then $x + z = p$

80. C

Sol. $g'(50) = \frac{1}{f'(2)}$

SECTION - B

81. 00004.00

Sol. Given polynomial is $(x^2 - 4x + 1)(x^2 - \lambda x + \mu)$ where $\lambda = -2$ and $\mu = a + 7, b = 2 - 4\mu = -26 - 4a$

$\therefore$ Roots are real $\therefore \mu \leq \frac{\lambda^2}{4}$

$\Rightarrow a + 7 \leq 1 \Rightarrow a \leq -6$ and $b \geq -2$

82. 00006.00

Sol. $f\left(e^{-\frac{i\pi}{3}}\right) = (a+c)\omega - b + c$; $\omega = -\frac{1+i\sqrt{3}}{2}$

$\Rightarrow (a+c)\left(-\frac{1}{2} + \frac{i\sqrt{3}}{2}\right) - b + c = 4038\sqrt{3}i - 4032$

$\Rightarrow a + c = 8076$; $\left(\frac{c-a}{2}\right) - b = -4032$

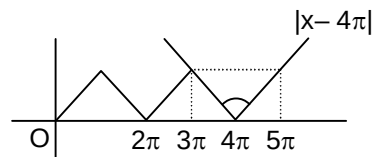
$\Rightarrow a = c = 4038$ and $b = 4032$

83. 00003.00

Sol. $\left[\sum_{r=0}^{\infty} \int_0^1 t^{4r} (1-t^2) dt = \int_0^1 (1-t^2) \frac{dt}{(1-t^4)} \right]$

84. 00015.00

Sol. $a = \sin^{-1} \sin(8\pi + 26 - 8\pi) = 26 - 8\pi$
 $b = \cos^{-1} \cos 36 = \cos^{-1} \cos(12\pi - (12\pi - 36)) = 12\pi - 36$
 $c = \cos^{-1} \cos(a + b) = \cos^{-1} \cos(4\pi - 10) = 4\pi - 10$
 $\Rightarrow |x - 4\pi| \leq \cos^{-1} \cos x$
 $\Rightarrow x \leq 15$



85. 10080.00

Sol. $f(x) = 10(x+1)(x+5)(x-2)(x-3)(x-4) + x^2 - x + 1$

86. 02021.50

Sol. $f(x) + f(2-x) = 1$

87. 01011.00

Sol. $\sin\left(3x - \frac{\pi}{6}\right) + \cos\left(2x + \frac{5\pi}{9}\right) = -2$
 $\Rightarrow \sin\left(3x - \frac{\pi}{6}\right) = -1$ and $\cos\left(2x + \frac{5\pi}{9}\right) = -1$
 $\therefore x = 2n\pi + \frac{11\pi}{9}; n \in \mathbb{Z}$

88. 00074.00

Sol. $\sum_{k=0}^7 k \cdot 2^k = 1538 \therefore 1538 + 8(n - 2^8 + 1) = 1898$
 $\Rightarrow n = 300 \therefore \text{Number of zeroes} = \text{exponent of 5 in } 300! = 74$

89. 01524.00

Sol. ${}^5C_1 \left[\frac{5!}{2!} + \frac{5!}{3!} \right] + {}^5C_2 4! = 640$

90. 00191.00

Sol. $\sum_{k=0}^{10} \frac{{}^{10}C_k {}^{10+k}C_{10}}{(-2)^k}$
Coefficient of x^{10} in $\left[{}^{10}C_0 (1+x)^{10} - {}^{10}C_1 \frac{(1+x)^{11}}{2} + {}^{10}C_2 \frac{(1+x)^{12}}{2^2} + \dots + {}^{10}C_{10} \frac{(1+x)^{20}}{2^{10}} \right]$
Coefficient of x^{10} in $(1+x)^{10} \left[{}^{10}C_0 - {}^{10}C_1 \left(\frac{1+x}{2} \right) + {}^{10}C_2 \left(\frac{1+x}{2} \right)^2 - {}^{10}C_3 \left(\frac{1+x}{2} \right)^3 + \dots - {}^{10}C_{10} \left(\frac{1+x}{2} \right)^{10} \right]$
Coefficient of x^{10} in $(1+x)^{10} \left(1 - \left(\frac{1+x}{2} \right) \right)^{10}$
Coefficient of x^{10} in $\frac{(1-x^2)^{10}}{2^{10}} = \frac{{}^{10}C_5}{2^{10}} = \frac{252}{1024} = \frac{63}{128}$